

Textured Relief Map

A reusable prompt template for QGIS, via an LLM with PyQGIS execution

Sections 1 to 9 are a single prompt. Fill in the values in double braces, attach your OpenTopography API key, and paste the whole thing into a session with a QGIS MCP plugin connected. Everything in those sections is written as instructions to the model, not to you. Section 10 is an appendix addressed to you rather than the model: a fill-in-the-blank message to send alongside this document, plus a short guide to choosing each value.

It produces one square relief map with a footer band carrying the title, a subtitle and an elevation legend. The style comes from stacking four raster layers rather than from any single hillshade, and most of the rules exist because the obvious approach fails in a specific way that is named alongside each one.

1. Efficiency rules, read first

- Run the entire pipeline as one PyQGIS script in a single execution call.** Do not step through download, reproject, derivatives, styling and layout as separate tool calls.
- Download the DEM exactly once.** Check whether the GeoTIFF already exists in the working folder and skip the download if it does.
- Rasters stay as GeoTIFF. Only vectors go in the GeoPackage.** Write derivatives as tiled, DEFLATE compressed GeoTIFFs: `-co TILED=YES -co COMPRESS=DEFLATE -co PREDICTOR=2`.
- Do not process at a higher resolution than the export needs.** Map item pixel width is `map width in mm / 25.4 * 300`. If the DEM is more than about 1.5 times that wide, resample it down first, then everything downstream runs on a smaller array.
- Two renders maximum.** One low resolution preview to confirm the composition, then the 300 dpi final.
- Return at most two images in the whole conversation.** Intermediate screenshots are the single most expensive thing you can do. No per layer previews, no before and after pairs.
- No color blindness simulation exports.** The palette rule in section 5 makes the result safe by construction, so a simulation pass adds nothing.
- Do not echo state.** No layer listings, no field dumps, no script printouts, no narrating each step. When finished, report only the output paths, the opacity values settled on, and anything changed from these instructions.
- Do not stop to ask questions mid run.** If something is ambiguous, choose sensibly, continue, and flag the choice at the end.
- Keep the project and the rasters.** A saved project means the next fix is a re-export rather than a re-run, which saves far more than deleting files ever will.

2. Data and extent

Use the OpenTopography API key in the attached file. Download the DEM for `{{AREA NAME}}`, extending past the feature itself so the surrounding landscape frames it. Do all work in `{{WORKING FOLDER}}`.

Extent, given as a center point and a half width

Requesting degrees by hand produces badly over wide frames. Deriving the box from a distance in kilometers is much harder to get wrong.

- Center `{{LAT}}`, `{{LON}}`, half width `{{N}}` km.
- Compute the box: `dlat = half_width_km / 110.9`, `dlon = half_width_km / (111.32 * cos(radians(lat)))`. Then `Ymin = lat - dlat`, `Ymax = lat + dlat`, `Xmin = lon - dlon`, `Xmax = lon + dlon`. Add about 5 percent padding so reprojection does not clip the edges.
- Size the half width to the visually interesting part, not to the landform's full footprint.** A volcano's gullied cone may be 12 km across while its ring plain runs 25 km. Frame the cone. As a rule, half width is roughly 1.2 times the radius of the interesting part.
- X is longitude and Y is latitude. Western longitudes are negative. API mapping: `west=Xmin`, `south=Ymin`, `east=Xmax`, `north=Ymax`, `outputFormat=GTiff`.
- Report the computed box and its size in kilometers before proceeding.

Choosing the DEM

Match resolution to the frame rather than reflexively grabbing the finest data available.

Required cell size is `frame width in metres / map item pixel width`. For a 220 mm map at 300 dpi that is about 2,600 px, so a 100 km frame needs roughly 40 m cells and a 20 km frame needs roughly 8 m. Finer data than that is wasted: it slows every raster pass and cannot be resolved on the page. Coarser data reads as soft, mushy terrain that no amount of vertical exaggeration fixes.

Dataset	When to use it
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COP30	Default. Copernicus 30 m, global, radar derived, void free. Correct for any frame wider than about 40 km. Clean data matters here because the fine hillshade layer amplifies noise.
USGS10m	US features framed under about 26 km, via OpenTopography's separate 3DEP endpoint. Parameter names differ from the global DEM API, so check that endpoint's own documentation.
CA_MRDEM_DTM, EU_DTM	Canada and Europe, when a true terrain model with no canopy step is wanted.
NASADEM	Only when canopy height is actively unwanted and no regional DTM exists. Covers 60N to 56S only, and is noisier than COP30.
National lidar	Where a country publishes a lidar DTM and the frame is small enough to need it. Expect a different vertical datum and projection from the global sets.

COP30 is a surface model, not a terrain model. Forest canopy appears as a real 20 to 30 m step, which can show as an abrupt edge at a park or field boundary. Sometimes striking, sometimes an unwanted artifact. Decide deliberately rather than discovering it in the final render.

If the required cell size is finer than anything available, say so and widen the frame rather than upsampling. Vertical datum for the global sets is EGM2008, orthometric metres. Apply no vertical conversion, and treat a z factor of 1 as the neutral baseline.

Projection

The download is EPSG:4326. Reproject to `{{PROJECTED CRS, metres, e.g. EPSG:32619}}` using bilinear resampling, and set the project CRS to match. Never run hillshade on unprojected degrees.

3. Derivatives

All three via gdaldem, inside the same script, written as compressed GeoTIFFs.

1. `hillshade_multi`, multidirectional, z factor `{{1.5 to 2.5, higher for low relief terrain}}`
2. `hillshade_micro`, azimuth 315, altitude 65, z factor `{{5 to 8}}`. This is what creates the fine etched filigree.
3. `slope`, in degrees

4. Layer stack

The stacking is the style. Do not collapse it into a single hillshade.

#	Layer	Symbology	Blend	Opacity
1	DEM	Singleband pseudocolor, hypsometric ramp, quantile spaced stops	Normal	100%
2	<code>hillshade_multi</code>	Pseudocolor, 0 to <code>{{SHADOW}}</code> , 255 to <code>{{ACCENT}}</code> , linear	Multiply	100%
3	<code>slope</code>	Pseudocolor, 0 degrees transparent, <code>{{35 to 45}}</code> degrees to <code>{{SHADOW}}</code>	Multiply	50 to 60%
4	<code>hillshade_micro</code>	Greyscale, stretched	Overlay	20 to 30%

5. Palette

Warm light against cool shadow, with lightness rising through every stop so the terrain reads from lightness alone. That is what makes it safe for color vision deficiency, and it is why no simulation pass is needed.

- Five stops: shadow `{{HEX}}` at L approximately 11, deep midtone `{{HEX}}` at L 26, midtone `{{HEX}}` at L 46, highlight `{{HEX}}` at L 70, accent `{{HEX}}` at L 91. L is CIELAB lightness, 0 to 100.
- Adjacent stops must differ by at least 12 lightness points.
- Stay within one or two adjacent hue families.
- **Never put green or olive against a warm highlight.** That is the pairing that collapses for red green deficiency. Blue against sand and blue against orange are both safe.

Put the ramp stops on quantiles, never on equal intervals

Position the five stops at the 2nd, 25th, 50th, 75th and 98th percentiles of the DEM's actual values. This is not optional, and it is the single most common way these maps fail. On a volcano, a linear minimum to maximum ramp puts the entire surrounding plain into the bottom 10 percent of the color range, and the frame becomes one flat dark mass with a bright dot in the middle. Quantile spacing gives the summit its highlight and keeps the lower terrain legible at the same time. Report the five elevation values the stops were placed at.

If the feature occupies a single narrow elevation band, compress the ramp to that band's own minimum and maximum rather than the full DEM range.

Flat water, if present

Extract cells at or below the water surface elevation as a polygon, store it in the GeoPackage, and symbolise it flat `{{VERY DARK HEX}}`, darker than any terrain shadow}} with a 0.3 mm shoreline stroke in the accent color. The shoreline then reads as a lightness edge, which keeps it legible under any color vision.

6. Layout

- **Page** `{{240}}` by `{{280}}` mm, a square map with a footer band beneath it, exported at 300 dpi.
- **Map item** `{{220 by 220}}` mm, 10 mm from the left, right and top edges.
- **The subject must occupy 50 to 70 percent of the frame width.** Below 40 percent the frame is too loose. Tighten the map item's extent and re-export, which costs one export rather than a re-run.
- **Page background** `{{DEEP HEX}}`, not white and not pure black. A deep desaturated version of the shadow hue at 8 to 18 percent lightness.
- **Map frame** 0.6 mm stroke in `{{FRAME HEX}}`, pale and high lightness, plus a second rectangle 2 mm outside it in the same color for a double border.
- **No coordinate grid or zebra border anywhere on the layout.** A grid whose CRS silently defaults to degrees while the interval is set in metres yields zero divisions and draws nothing, with no error raised. A linked scale bar is one object, is self validating, and cannot fail silently in the same way.

Scale bar

A standard scale bar item, placed inside the map at bottom left.

```
from qgis.core import QgsLayoutItemScaleBar, QgsUnitTypes, QgsLayoutPoint
from qgis.core import QgsUnitTypes as U
from qgis.PyQt.QtGui import QColor, QFont

sb = QgsLayoutItemScaleBar(layout)
sb.setLinkedMap(map_item)          # must be set BEFORE sizing, or the bar computes nothing
sb.setStyle('Single_Box')
sb.setUnits(QgsUnitTypes.DistanceKilometers)
sb.setUnitLabel('km')
sb.setUnitsPerSegment({{2}})       # pick so the whole bar spans 15 to 20% of map width
sb.setNumberOfSegments(4)
sb.setNumberOfSegmentsLeft(0)
sb.setHeight(3.0)
sb.setFillColor(QColor('{{FRAME HEX}}'))
sb.setFillColor2(QColor('{{BACKGROUND HEX}}'))
sb.setLineColor(QColor('{{FRAME HEX}}'))
sb.setLineWidth(0.3)
sb.setFont(QFont('{{FONT}}', 11))  # letter spacing 0
sb.setFontColor(QColor('{{FRAME HEX}}'))

# dark backing so it stays readable over pale terrain
sb.setBackgroundEnabled(True)
bg = QColor('{{BACKGROUND HEX}}'); bg.setAlphaF(0.65)
sb.setBackgroundColor(bg)

layout.addLayoutItem(sb)
# 8 mm in from the map's left and bottom inner edges
sb.attemptMove(QgsLayoutPoint(x_mm, y_mm, U.LayoutMillimeters))
```

Verify before exporting: the bar must show labelled segments carrying a real distance. If it renders as an empty box or at zero width, `setLinkedMap` did not take. Total bar length should be roughly 15 to 20 percent of the map width, so adjust `unitsPerSegment` until it is.

North arrow

- Total height 16 mm for the arrow itself. A small decorative arrow does not register at this page size.

- A vertical shaft 1.4 mm thick with a solid filled triangular head 5 mm wide by 7 mm tall, in `{{FRAME HEX}}`.
- Label **N** directly above it, bold, 16 pt, same color.
- **Position the whole assembly so the top of the N sits at least 8 mm below the map's inner frame line** and 10 mm in from the right edge. Measure the text's rendered height and offset from that, rather than inseting only the arrow graphic, or the N will touch the frame.
- Give both the arrow and the N a 0.6 mm halo in the background color so they stay readable where the terrain beneath is pale.

Text rendering rules, applying to every text item

Set letter spacing to 0 and never change it. QGIS letter spacing is specified in points, so a value that sounds modest against a 34 pt title is enormous against an 8 pt label. The failure mode is that text stretches past its frame and wraps to one character per line. Do not use `QFont.setLetterSpacing`. If a title should feel wide, use a larger size or heavier weight instead.

- Do not enable justified alignment, auto fit, or any resize to fit frame option. Size each text frame generously and let the text sit inside at its natural width.
- After building the layout, check every text item's rendered bounding box against its frame. Fix anything wider than its frame or wrapped onto an unexpected second line before exporting.

Title block, in the footer band

- Title `{{TITLE}}`, 34 pt bold, letter spacing 0, in `{{FRAME HEX}}`, left aligned to the map's left edge and sitting about 12 mm below the frame.
- Subtitle directly beneath at 9 pt, in `{{FRAME HEX}}` at 70 percent opacity: `{{feature type, region, country, center coordinates, DEM source, projection}}`.
- **Verify place names against current official sources**, not just against coordinates. Park and feature names change by legislation and a coordinate check will not catch it. If the map credits a national mapping agency, use that agency's current gazetteer.

Elevation legend, bottom right of the footer band

- Aligned to the map's right edge and vertically centered against the title.
- A horizontal gradient bar 75 by 8 mm using the exact same color stops as the DEM ramp, with a 0.2 mm stroke in `{{FRAME HEX}}`.
- Labels beneath at 11 pt: minimum elevation at the left end and maximum at the right end, both with m units, pulled from the actual DEM statistics.
- The word ELEVATION above the bar at 10 pt, letter spacing 0, left aligned to the bar's left edge. Not centered over it, and not stretched to its width.
- If the source DEM reports heights above an ellipsoid rather than a geoid, do not label the legend as elevation above sea level. Either state the datum or label the ends Lower and Higher with no numbers.

No scale caption text, since the scale bar carries it. No other text, no legend frame, no logo, no graticule labels, no title box. The terrain is the composition and the footer is a caption strip. Export to PNG at 300 dpi.

7. Preview checklist

Check the preview against every item below before the final export. Each of these has failed silently, and a layout renders successfully with all of them broken.

1. The scale bar sits inside the map at bottom left, with readable segment labels and a real distance on them.
2. The title reads as one continuous word, not spaced out letters spanning the page.
3. ELEVATION appears as a whole word above the legend bar.
4. The N of the north arrow has clear space below the map frame and is not touching it.
5. Nothing overlaps. Title, subtitle and legend each occupy their own space in the footer band.
6. There is no coordinate grid or zebra border anywhere on the layout.
7. The subject fills 50 to 70 percent of the frame width. If it looks small and lost, tighten the extent.
8. No more than about 30 percent of the frame sits in the darkest ramp stop. A large flat dark area means the stops are not on quantiles.

Fix anything that fails and re-check the preview. Only then export the final.

8. Outputs

In the root of `{{WORKING FOLDER}}` :

1. `{{name}}_relief.png`, the final 300 dpi layout
2. `{{name}}.qgz`, the saved project with all styling and the finished layout intact
3. `{{name}}.gpkg`, only if there is vector data. If the feature has no water polygon or other vectors, say so and skip the file rather than writing an empty one. State explicitly what went into it, since a large GeoPackage on a feature with no vectors means something unintended was written, most likely a raster.

In a `_data` subfolder keep the rasters: the reprojected DEM, both hillshades, the slope raster, and their `.aux.xml` sidecars. Tiled and compressed these are small, and the project needs them in order to reopen.

Save the project after all styling and the layout are complete, not partway through, and set relative paths first or the project breaks the moment the folder moves.

```
project.writeEntryBool('Paths', '/Absolute', False)
project.write(os.path.join(out_dir, '{{name}}.qgz'))
```

Afterwards, confirm the file exists and is non zero and report its path. Do not assume `write()` succeeded.

Delete only the scratch folder, the raw EPSG:4326 download once reprojection is done, any preview renders, and any clipped or resampled intermediates that nothing references. Nothing else. On Windows, GDAL holds a read lock on any raster still loaded in the project, so `os.remove` will throw `PermissionError` or fail silently inside a try block. Remove the layer from the project and force garbage collection first.

9. Troubleshooting

Symptom	Cause and fix
Flat and muddy	Layer 2's ramp covers too narrow a value range. Push the shadow darker and the accent lighter before touching any opacity.
Airbrushed, no texture	Raise <code>hillshade_micro</code> 's z factor or its opacity. That layer is what makes the terrain look engraved.
One dark mass with a bright spot	Ramp stops are on equal intervals instead of quantiles. See section 5.
Subject looks small and lost	Half width was sized to the landform's full footprint rather than the interesting part. This is a layout property, so it is a re-export and not a re-run.
Text wrapped one character per line	Letter spacing applied in the wrong unit. Set it to 0 everywhere.
Scale bar renders empty	<code>setLinkedMap</code> was called after sizing, or not at all.
Unsure the palette is safe	Convert the exported PNG to greyscale. Hillshading already encodes shape as lightness, so if the terrain still reads in greyscale it is safe for every type of color vision deficiency. This takes seconds and needs no extra render.

10. Appendix, the kickoff message

This section is addressed to the person running the template, not to the model. Send the message below in the same turn as this document and the API key file. Fill the double brace fields, and delete the three optional blocks outright when they do not apply rather than leaving them in with empty values.

Go ahead and make me the **{{FEATURE NAME}}** map in QGIS. Instructions are in the attached template. Follow the efficiency rules, the text rendering rules and the preview checklist strictly, save the `.qgz` project, and keep the rasters in a `_data` subfolder. My OpenTopography API key is in the attached txt file.

Extent: center `{{LAT}}`, `{{LON}}`, half width **{{N}} km**. Compute the bounding box, add about 5 percent padding, and report the box and its size in km.

DEM: `{{DATASET}}`. At a roughly `{{X}}` km frame the required cell size is about `{{Y}}` m, so this resolution is appropriate. Do not substitute finer data.

Projected CRS: `{{EPSG}}`, bilinear resampling.

Optional, include only when upsampling more than about 2 times: A `{{X}}` km frame needs about `{{Y}}` m cells and this DEM is `{{Z}}` m, so we are upsampling about `{{N}}` times. After reprojecting, resample to `{{10}}` m using cubic spline before computing derivatives. It adds no information but prevents staircase artifacts.

Z factors: `{{A}}` for the multidirectional hillshade and `{{B}}` for the micro hillshade.

Elevation ramp: place the five palette stops at the 2nd, 25th, 50th, 75th and 98th percentiles of the DEM's actual values, not at equal intervals. Report the five elevation values used.

Palette `{{NAME}}`: lightness rising through every stop: `{{HEX}}` at L 12, `{{HEX}}` at L 26, `{{HEX}}` at L 45, `{{HEX}}` at L 68, `{{HEX}}` at L 92. Page background `{{HEX}}`, frame and text `{{HEX}}`.

Optional, include only when the frame contains a lake, reservoir or sea: Extract cells at or below about `{{E}}` m as a polygon, keeping only the connected body containing `{{seed lat, lon}}`. Store it in the GeoPackage as `{{name}}`, style it flat `{{VERY DARK HEX}}` with a 0.3 mm shoreline stroke in `{{ACCENT HEX}}`, above the raster stack. Report its area in square kilometers so I can sanity check it. Any island inside it must keep its full terrain treatment, so the polygon needs a hole rather than a cover.

Optional, include only when labels are wanted: Labels, `{{N}}` of them, Segoe UI 10 pt, `{{FRAME HEX}}`, 0.8 mm dark halo in the background color, names only. Snap each to the nearest local elevation maximum within `{{800}}` m: `{{Name}}` near `{{lat, lon}}`, and so on. Offset them so none overlaps a crest or another label. Label nothing else.

Typography: Segoe UI throughout. Title `{{TITLE}}` at 42 pt bold. Reduce the size if it will not fit on one line. Do not wrap it and do not apply letter spacing. Subtitle at 13 pt: `{{feature type, region, country, center coordinates, DEM name, CRS}}`. Letter spacing 0 on every text item.

Scale bar bottom left inside the map, 4 segments of `{{S}}`. No coordinate grid and no each block caption. The N of the north arrow must clear the map's inner frame line by at least 8 mm.

Optional, include only when there is no water and no labels: There is no vector data in this run, so skip the GeoPackage entirely and tell me that is what you did.

Save as `{{name}}_relief.png`, `{{name}}.qgz` and `{{name}}.gpkg` in `{{WORKING FOLDER}}`.

Filling the fields

Field	How to choose
Half width	About 1.2 times the radius of the visually interesting part, not the landform's footprint. The subject should end up filling 50 to 70 percent of the frame width.
DEM	Required cell size is frame width divided by 2,600. <code>COP30</code> for frames wider than about 40 km, <code>USGS10m</code> for US frames under about 26 km, a national lidar DTM warped straight to the target resolution where one exists, <code>CA_MRDEM_DTM</code> or <code>EU_DTM</code> for bare earth regional coverage.
Projected CRS	The UTM zone for the center longitude, or a national grid where one fits the area better and avoids a zone boundary.
Z factors	Elevation range under 500 m, use 2.5 and 8. From 500 to 1,500 m, use 1.8 and 6. From 1,500 to 3,000 m, use 1.5 and 5. Over 3,000 m, use 1.2 and 4. When upsampling more than 2 times, cap the micro value at 4.0.
Palette	Five stops at lightness of about 12, 26, 45, 68 and 92, with adjacent stops at least 12 apart, inside one or two adjacent hue families. Never green or olive against a warm highlight.
Scale bar segment	About frame width times 0.045, rounded to 1, 2, 5 or 10 in km or m.

Textured Relief Map prompt template. Written for QGIS driven through an MCP plugin with PyQGIS execution. Elevation data through the OpenTopography API. Adapt the placeholder values freely.